

Arduino Uno Starter Set Kit

WHAT'S IN THE KIT

- 12 Elegoo Arduino Uno starter sets with components, sensors, and project guides



QUICK-START GUIDE

1. Install the Arduino IDE (or use the web editor).
2. Each Elegoo set includes a guide — start with Lesson 1: Blink.
3. Progress through LEDs → buttons → sensors → displays.
4. Pairs work well: one drives the wiring diagram, one builds.

FULL LESSONS IN THIS GUIDE

GRADES 6-8	Traffic Light City	page 2
GRADES 9-12	Design-a-Device	page 3

PRO TIP: The included Elegoo lesson PDFs are genuinely good — let students self-pace through them.

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LESSON · GRADES 6-8

Traffic Light City

LEARNING OUTCOMES

- Students will wire and program LEDs with timed logic
- Students will coordinate multiple outputs into a system
- Students will trace a program's timing behavior

MATERIALS

- Elegoo Arduino starter sets (1 per pair)
- Laptops with Arduino IDE (or web editor)
- Elegoo lesson guide (included PDFs)
- Masking tape roads for the 'city grid'

INSTRUCTIONS

1. Pairs complete the Blink lesson to verify their toolchain.
2. Build the 3-LED traffic light following the kit guide.
3. Program the real sequence: green → yellow → red with correct timing.
4. Pairs join tables into an intersection; lights must coordinate so traffic never collides.
5. Stress test: teacher 'drives' a toy car through; any crash means retiming.
6. Reflection: how did you keep two independent programs in sync?

ACTIVITY

Traffic Light City — working traffic lights that must coordinate across an intersection. Synchronizing separate Arduinos surfaces timing concepts no worksheet can.

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LESSON · GRADES 9-12

Design-a-Device

LEARNING OUTCOMES

- Students will translate a real-world need into device requirements
- Students will build a functional prototype integrating sensors and outputs
- Students will pitch a product with a live technical demo

MATERIALS

- Elegoo Arduino starter sets
- Laptops with Arduino IDE
- Prototyping materials (cardboard, tape, rubber bands)
- Pitch rubric: problem, demo, cost, next steps

INSTRUCTIONS

1. Ideation: list small real annoyances at school or home; pick one per team.
2. Write requirements: what does the device sense, decide, and do?
3. Component check: confirm the kit can support the design before committing.
4. Build sprint across 2-3 sessions; keep a build log.
5. Dry-run demos with a partner team; fix the most embarrassing failure.
6. Demo day: 3-minute pitches with the working device, judged on the rubric.

ACTIVITY

Design-a-Device — from annoyance to working prototype: plant-thirst monitors, door alarms, posture buzzers, study timers. The live demo requirement keeps designs honest.